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Grayscale Image-based Deep Learning Model for Efficient and Scalable Malware Detection

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***Abstract*—Malware generally implies a significant threat of information security regarding to jeopardize user data, privacy, and assets. Therefore, early detection is vital to prevent serious consequences. Moreover, the diverse array of carriers and technologies that have created a disconnect between training scenarios and real-world detection methods. This illustration actually emphasizes this gap. Because the identification of new malware variants mostly hinders threat responses. Therefore, researchers are prone to re-think to solve such effects. Image-based malware classification could be an authentic option to detect new and evolving malware variants. In this work an attempt has been attributed introducing an innovative deep learning-based detection method that classifies malware image in the form of grayscale. In this study, the MalImg dataset has been taken upon and evaluated a model using key performing metrics to achieve remarkable accuracy compared to existing benchmarks. Further, it also establishes a robust system for real-time malware classification paving the way for effective visual detection. Additionally, it suggests that computers can autonomously identify malware threats that mark a potential shift enhancing security measures for complex digital landscape.**

***Index Terms*—Malware Detection System, Convolutional Neural Network (CNN), Cyber Security, Deep Learning**

I. Introduction

In the age of Industry 5.0, the applications of internet-enabled embedded devices have increased rapidly across various domains. Prarally, Cyber attackers are making well-coordinated, far more sophisticated attacks than anything experienced before [1]. Despite the enormous breakthroughs in security systems and their continuous expansion, malware remains a major threat in the cyber world. [2]. The term malware represents malicious software. It is a program installed without a user's authorisation. It performs many unwanted activities, like mining information, ruining the system's performance, leveraging network weaknesses, hijacking a device, Etc. The most common types of malware are:: Viruses: Capable of spreading to other systems by replicating themselves and requiring user input, Worms: Which spread automatically without user input, Adware: Hit the user with unauthorised advertisements (ads), Spyware: They automatically download into a system and begins secretly watching your internet activity without your knowledge or consent, Ransomware: Locks or encrypts a system and demands payment for it, Bots: A group of machines carrying the same infection, Rootkits: Gives the hacker access to the infected machine and hides its existence, Trojan Horses: recognise itself as a regular program to bluff users to install it on their systems, Bugs, etc. [3], [4]. Every day, we are introducing more and more unknown viruses. A report says McAfee encounters an average of 1.1 million such programs and potentially unwanted apps (PUA) daily, and it is very much alarming for the world of the internet. Malware is being produced at an ever-increasing rate with the help of different software or techniques. In a recent research, the AV-TEST Institute stated that more than 4.5 lakh new potentially unwanted applications and malicious software are discovered daily. Modern, cutting-edge malware detection techniques focus significantly on antivirus programs. They detect malicious programs based on signature patterns. This software has several limitations, including the inability to detect encrypted or packed malware. Using specific packers to reuse the code might create a new malware strain. Thus, signature-based technique overlooks such types of programs, and they are undetected for an extended period of time. Basically, three approaches are followed to analyse malware: static, dynamic, and reverse engineering. Static analysis is a way of analysing code without execution. In contrast, dynamic analysis is performed during runtime. In reverse engineering, the functionality and impact on the system of the code are verified by converting binary instructions into high-level constructs. A considerable amount of research has been conducted in this domain. Before code obfuscation, static analysis was not a difficult task. In dynamic analysis, behaviour of the data is analysed in a sandbox, such as CWSandbox [5] and TTAnalyzer [6], and thus it is not affected by code obfuscations. However, dynamic analysis just simulates the program for a short period, and occasionally it cannot trigger the entire execution of the program [7]. Malicious techniques give false alarms for malware and consider malicious behaviour normal. Current approaches can only detect known risks; they cannot detect malware that has already been introduced into the data or new threats it may produce. In order to identify malware, the researchers recommended foundational AI approaches like ML and DL [8]. The key contributions are

- Identify the gaps in the existing literature.
- Analysis of MalImg dataset according to the malware Family.

- Visualising a malware image and then training a classifier based on DL techniques to improve performance.
- Comparison with present research works.

This work is organised as follows: Section II is about related works and their contributions. The System model is discussed in section III. Results and analysis are in Section IV. conclusion is done in sections V.

II. Related Work

This section highlights different research directions in this domain. In 2010, Nataraj et al. [9] presented a method of visualizing malware. A malware binary can be converted into a two-dimensional array and visualized in the form of a grayscale image. This opens a new way for malicious analysis. Yang et al. [10] proposed a method that amalgamates frequency domain techniques with feature alignment to improve variant malware detection. This approach reduces distribution differences between labelled samples and new samples. By using malware image and the discrete cosine transform (DCT) for feature extraction, the model leverages a deep residual network to align features effectively across both domains. While their current experiments rely on specific public datasets that represent common malware families, they do not cover the other malware variations found in real-world scenarios. Sun et al. [7] proposed CNN and RNN-based models to identify malware families. Malware detectors use two inputs. First is the knowledge to recognize the program's harmfulness, and the other is the program that needs to be tested. Conti et al. [11] noted that when a new malware family emerges, typical deep learning classifiers must be retrained, which poses a significant challenge. They proposed a visualization technique that represents binary malware into an image of a 3-channel to classify malware. Nevertheless, they needed to work on dynamic analysis. To visualize malware binaries, Xiao et al. [12] used structural entropy. They then extracted the features using a CNN and employed SVM for classification. Son et al. [13] also worked on image-based malware classification, where the input was of low dimension. They utilized the Malimg and Malheurv datasets and performed classification. Wang et al. [14] suggested a hybrid analysis of Android malware by augmenting fuzzing. Suarez et al. [15] also worked on Android malware detection and the frequency of harmful functionality was tracked over time. They took 1.2M samples that spanned over a long period and analyzed rider behaviours or methods from all observed malware. Nike et al. [16] proposed a static malware analysis approach that utilizes fuzzy and import hashing techniques. However, they didn't employ other analysis techniques, such as dynamic and reverse engineering. In the domain of Industrial IoT, malware is also a significant challenge today. Esmaeili et al. [17] worked on IIoT-based malware detection. They examined query-based black-box malware on IIoT data and stateful detection of these attacks. Torabi et al. [18] proposed signature-based/static analysis techniques to detect malware in IoT environments. According to their findings, the old executable codes are used to create new threads. A CNN-based model was presented by Kumar et al. [19] and tested on Microsoft BIG and MalImg data.

III. System model

A. Data acquisition

In this work, the MalImg dataset is used for performance evaluation. This dataset contains 9,339 malware sample from one of the twenty-five malware classes and families. Table I shows the family, class, and total count of different malware. Among these, a larger number of samples are from Allaple. A family from Skintrim.N has the lowest number of samples. These samples are represented in the form of greyscale images. Additionally, the collection contains a variety of samples from different malware families. Figure 2 shows the class distribution of each malware family in this dataset.

TABLE I
MalImg Datase

Class	Family	Count
Backdoor	Rbot!gen	158
	Agent.FYI	116
Dialer	Instantaccess	431
	Dialplatform.B	177
	Adialer.C	125
Password Sealer	Lolyda.AA1	213
	Lolyda.AA2	184
	Lolyda.AT	159
	Lolyda.AA3	123
Rogue	Fakerean	381
TDownloader	Dontovo. A	162
	Obfuscator. AD	142
	Swizzot. gen!I	132
	Swizzor. gen!E	128
	Wintrim. BX	97
	Wintrim. BX	97
Trojan	C2Lop. gen!g	200
	Alueron.gen!J	198
	C2Lop.P	146
	Malex.gen!J	136
	Skintrim.N	80
	Skintrim. N	80
Worm	Allaple. A	2949
	Allaple. L	1591
	Yuner. A	800
	VB. AT	408
Worm:AutoIT	Autorum.K	106

Performed data visualisation on different types of malwares in grayscale image and presented in Figure 2.

B. Data Preprocessing

Before using data, preparation is a necessary task, and it also increases the model's performance. In this phase, we performed multiple operations on data(like cleaning, integration, transformation, reduction, discretization, Etc.) and make an appropriate format for the model.

1) Data Deduplication: Data deduplication is a process for getting rid of redundant copies of repeated information. It helps in better training and performance of the model.

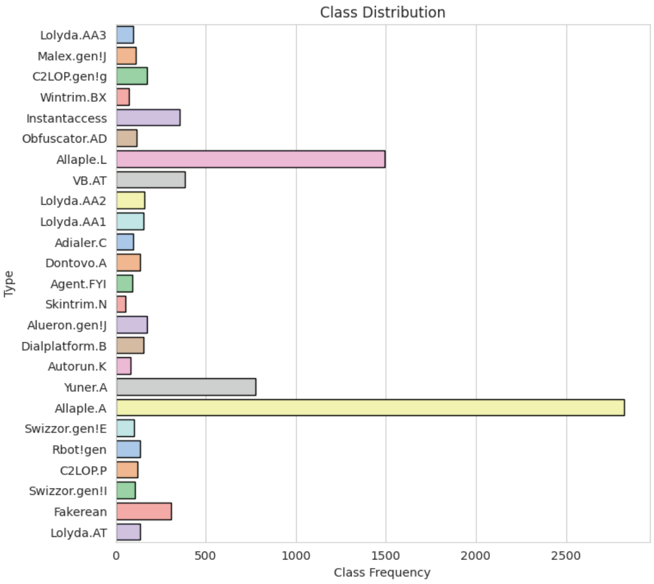

Fig. 1. Malware Family in MalImg dataset

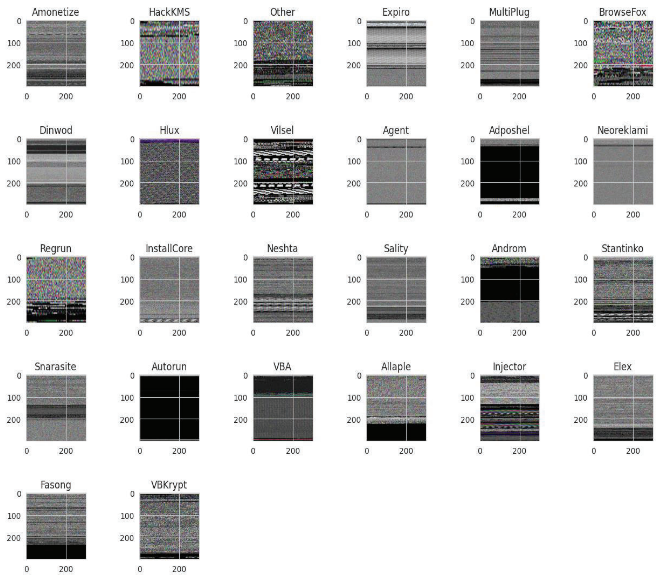

Fig. 3. Sample Malwares of different types in grayscale_MaleVis Dataset

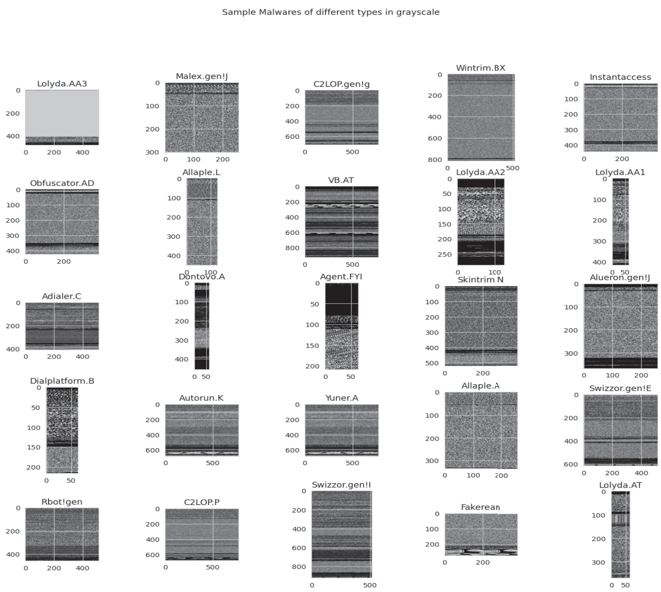

Fig. 2. Sample Malwares of different types in grayscale_MalImg Dataset

C. *Convolutional neural network*

It is one type of mighty Deep Neural Network primarily used in computer vision. It enables a computer to comprehend and analyze visual information or images. Here, the CNN model is utilised for malware image classification. The images are in the form of a greyscale. The first three layers of this network are convolutional, while the latter two levels are entirely linked. A malicious program is input over the network as a greyscale picture. The network outputs the predicted class of the given malware sample. Figure 4 demonstrates the proposed model and its architecture.

IV. Results and Analysis

A. *Performance Metrics*

The performance of the model is measured in term of performance metrics as follows:
Accuracy: It is a ratio between correct predictions and total predictions. It defines as

$$Accuracy = \frac{T_P + T_N}{T_P + F_P + T_N + F_N} \tag{1}$$

Precision: It provides information on the percentage of positive identifications that were accurate.

$$P_r = \frac{T_P}{T_P + F_P} \tag{2}$$

Recall: It tells about the proportion of actual positives that are identified accurately. It is the ratio of all data that are correctly classified as Attacks to all the data that are actually Attacks. It is also known as Detection Rate.

$$R_e = \frac{T_P}{T_P + F_N} \tag{3}$$

F_1- score: The harmonic mean is a way to balance between precision and recall.

$$F_1 = 2 * \frac{P_r * R_e}{P_r + R_e} \tag{4}$$

Figure 5 presents the two losses over epochs to visualise the model training and generalisation to unseen data. The decreasing graph (blue line) shows that the model was trained well in a stable manner. Figure 6 shows that the model is performing well, and the highest accuracy is calculated as 98.53%. AUC graph between training and validataion is presented in Figure 7. Figure 8, Figure 9, Figure 10, are presenting different graphs on False positive, Precision, Recall.

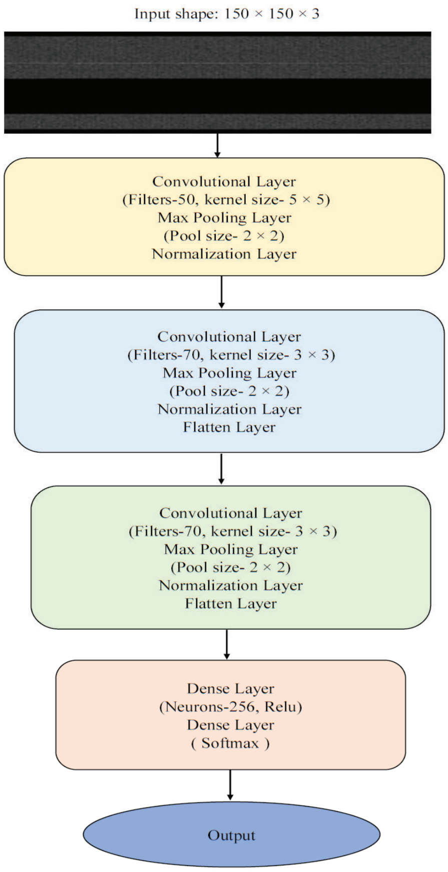

Fig. 4. Proposed CNN Model

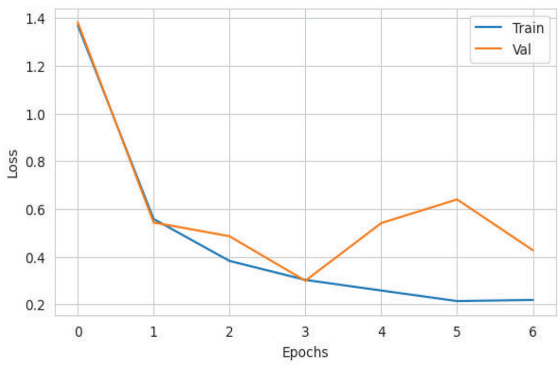

Fig. 5. Training vs Validation Loss

A detailed comparison has been conducted on various performance metrics between some existing models and this proposed model, as presented in Table II. The proposed CNN models demonstrated highly robust performance, achieving 98.53% accuracy, 96.55% precision, 97.01% recall, and an F1-score of 96.77%. These metrics indicate that our proposed model not only surpasses the performance of the existing literature but also demonstrates superior capabilities compared to the base model in the critical task of classifying benign

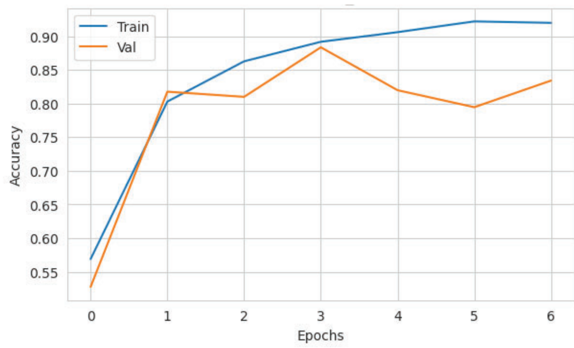

Fig. 6. Training Va Validation Accuracy

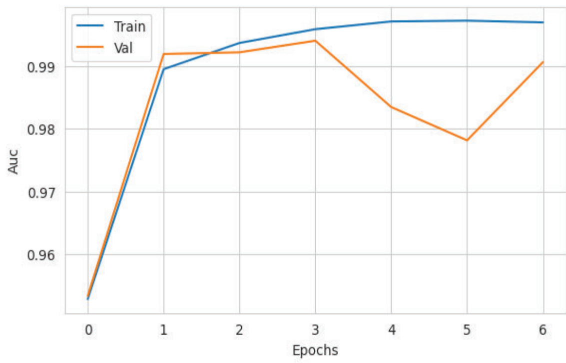

Fig. 7. Training Va Validation AUC

and brute-force attacks. To further illustrate these findings, different graphs are presented. In this visual representation, the blue bar represents the performance of the training, while the orange bar represents the validation results. This stark contrast emphasizes the efficacy and robustness of our model. Moreover, our thorough analysis reinforces the conclusion that the proposed CNN model not only outperforms efficiently but also establishes a higher benchmark for future research in the field. These results not only highlight the potential of this approach but also contribute significantly to advancing the

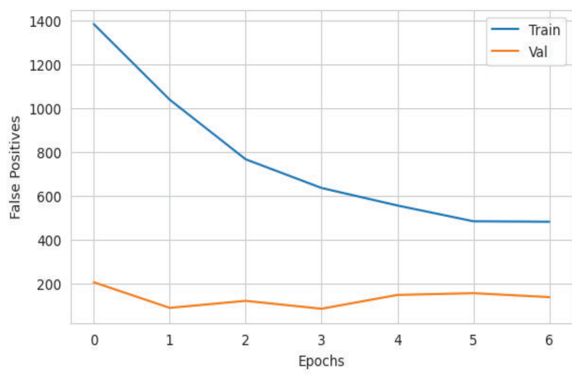

Fig. 8. Training Va Validation False positive

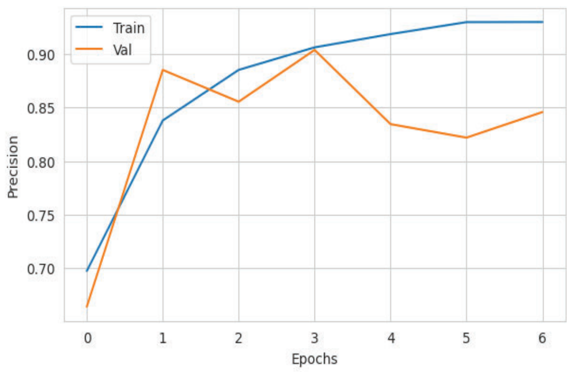

Fig. 9. Training Va Validation Precision

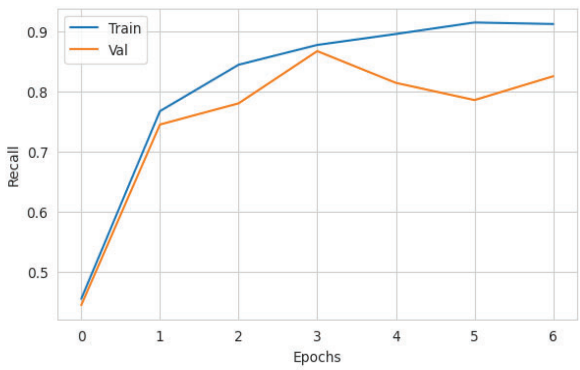

Fig. 10. Training Va Validation Recall

TABLE II
PERFORMANCE COMPARISON

Ref.	Dataset	Algorithm	Accuracy	Precision	Recall
[20]	R2-D2	CNN	95.7	96	96
[21]	Self created	Deep Neural Networks, Droid-Malware Detector	90	91	91
[22]	Drebin, Genome, Virusshare, and VirusTotal	SLR	97.58	96.24	96.2
[25]	MalImg	Feature-Alignment, domain adaptation	92.93	93.74	92.94
[23]	Android	Graph Convolutional Nets	92.30	91.50	93.30
[24]	MalImg	CNN	97.28	96.23	96.07
Proposed Model	MalImg	CNN	98.53	96.55	97.01

state of knowledge in cyber security attack classification.

B. Scalability

To assess the scalability of the proposed approach, this study expands beyond the Malimg dataset by integrating the MaleVis dataset [26]. Malimg consists of approximately 9,339 grayscale samples from 25 families, while MaleVis includes 14,226 RGB images from 26 classes. Figure 3 presents the malware sample present in the MaleVis dataset. The experiment conducted with the new dataset achieved an accuracy of 96.3%. The result confirms the model's capability to scale effectively from smaller to larger datasets, highlighting its reliability for cybersecurity applications.

V. Conclusion and Future Work

Numerous studies have been conducted to detect malicious programs in various scenarios. In this paper, first performed reviewed existing literature and, after careful analysis, proposed a simple, efficient, and effective technique for classifying malware based on images using deep learning. These images represent malware binaries transformed into a two-dimensional array. We successfully implemented a CNN approach to identify both new and modified malware images within a class of known malware. Our experiments utilized the benchmark MalImg dataset, achieving accuracy, precision, recall, and F1-score of 98.53%, 96.55%, 97.01%, and 96.77%, respectively. Compared to existing methods, our proposed model demonstrates promise for effective malware classification. By integrating the MaleVis dataset, the scalability of the model is also tested. We intend to utilise larger real-time datasets in the domain of Industrial Internet of Things (IIoT) and other domains in the future.

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